Experiment 3

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LO4: Containerize, orchestrate and deploy applications using Docker, Docker Compose, Kubernetes, and Cloud Infrastructure Tools.

Aim: To deploy multi-service applications using Docker Compose.

What is Docker Compose?

Docker Compose is a tool used to define, configure, and run multi-container Docker applications. Instead of starting each container individually, Docker Compose allows you to manage all the services of an application using a single configuration file called compose.yaml (or docker-compose.yml in older versions).

It simplifies the deployment of applications that require multiple containers, such as a web server, database, cache, and message broker.

Why Use Docker Compose?

- Simplifies the deployment of multi-container applications.
- Allows all services to be defined in a single YAML file.
- Automatically creates a network so services can communicate.
- Makes applications portable and reproducible across different systems.
- Supports scaling services and managing container dependencies.
- Reduces manual configuration and repetitive commands.

Key Components of Docker Compose

1. Services

A **service** defines a container that runs a specific part of the application, such as:

- Web server (Flask, Django, Node.js)
- Database (MySQL, PostgreSQL)
- Cache (Redis)

Each service can specify:

- Docker image or build context
- Port mappings
- Environment variables
- Volumes
- Networks

2. Images

An **image** is a template used to create containers. Docker Compose can:

- Pull images from Docker Hub (e.g., mysql:8.0)
- Build custom images using a Dockerfile

3. Containers

A **container** is a running instance of an image. Docker Compose automatically creates and manages containers for all defined services.

4. Networks

Docker Compose creates a default network for the application, allowing containers to

communicate using their service names instead of IP addresses.

Example:

- web connects to db using the hostname db.

5. Volumes

Volumes provide persistent storage so that data remains available even if containers are stopped or recreated.

Example:

- MySQL stores database files in a volume to prevent data loss.

Docker Compose Workflow

1. Create the application source code.
2. Write a Dockerfile for custom services.
3. Create a compose.yaml file defining all services.
4. Build the application images.
5. Start all services using Docker Compose.
6. Access the application through the exposed ports.
7. Monitor logs and manage containers.
8. Stop and remove services when they are no longer needed.

Advantages of Docker Compose

- Easy deployment of multi-container applications.
- Single configuration file for all services.
- Automatic network creation and service discovery.
- Simplified container management.
- Supports environment variables and persistent storage.
- Makes development, testing, and deployment more consistent.

Limitations of Docker Compose

- Best suited for development, testing, and small-scale deployments.
- Not intended for large-scale production orchestration.
- Limited advanced features compared to container orchestration platforms such as Kubernetes.

Common Docker Compose Commands

Command	Description
docker compose up	Creates and starts all services.
docker compose up -d	Starts services in detached (background) mode.
docker compose build	Builds images for services defined in the Compose file.
docker compose ps	Lists running containers managed by Docker Compose.
docker compose logs	Displays logs from all services.
docker compose restart	Restarts running services.
docker compose stop	Stops services without removing them.
docker compose start	Starts previously stopped services.
docker compose exec <service> <command>	Runs a command inside a running service container.

Command	Description
docker compose down	Stops and removes containers, networks, and other resources created by Compose.

Contents of .yml:

```
version: '3.9'

services:
  web:
    build: .
    ports:
      - "5000:5000"
    depends_on:
      - db

  db:
    image: mysql:8.0
    environment:
      MYSQL_ROOT_PASSWORD: root
      MYSQL_DATABASE: mydb
    volumes:
      - mysql_data:/var/lib/mysql

volumes:
  mysql_data:
```

Conclusion

Docker Compose is an efficient tool for deploying and managing multi-service applications. By defining services, networks, and volumes in a single configuration file, it automates container deployment, simplifies application management, and ensures consistent environments across development and testing. It is widely used for applications that require multiple interconnected services, reducing setup time and improving portability.

```
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\L15>docker --version
Docker version 29.2.1, build a5c7197

C:\Users\L15>docker compose version
Docker Compose version v5.0.2

C:\Users\L15>mkdir my-multi-app

C:\Users\L15>cd my-multi-app

C:\Users\L15\my-multi-app>mkdir web

C:\Users\L15\my-multi-app>touch docker-compose.yml
'touch' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\L15\my-multi-app>type nul > docker-compose.yml

C:\Users\L15\my-multi-app>type nul > web\Dockerfile

C:\Users\L15\my-multi-app>type nul > web\app.py

C:\Users\L15\my-multi-app>type nul > web\requirements.txt

C:\Users\L15\my-multi-app>cd web

C:\Users\L15\my-multi-app\web>cd Dockerfile
The directory name is invalid.

C:\Users\L15\my-multi-app\web>cd dockerfile
The directory name is invalid.

C:\Users\L15\my-multi-app\web>cd Dockerfile\
The directory name is invalid.

C:\Users\L15\my-multi-app\web>cd Dockerfile
The directory name is invalid.

C:\Users\L15\my-multi-app\web>cd ..

C:\Users\L15\my-multi-app>notepad web\Dockerfile

C:\Users\L15\my-multi-app>code .

C:\Users\L15\my-multi-app>docker compose up -d --build
empty compose file

C:\Users\L15\my-multi-app>docker compose up -d --build

C:\Users\L15\my-multi-app>docker compose up -d --build
time="2026-07-16T19:40:18+05:30" level=warning msg="C:\\Users\\L15\\my-multi-app\\docker-compose.yml:
the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
"
unable to get image 'my-multi-app-web': error during connect: Get "http://%2F%2F.%2Fpipe%2FdockerDeskt
opLinuxEngine/v1.51/images/my-multi-app-web/json": open //./pipe/dockerDesktopLinuxEngine: The system
cannot find the file specified.

C:\Users\L15\my-multi-app>
```

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

Loading personal and system profiles took 1975ms.
(base) PS C:\Users\L15> docker version
Client:
 Version:           29.2.1
 API version:       1.53
 Go version:        go1.25.6
 Git commit:        a5c7197
 Built:             Mon Feb  2 17:20:16 2026
 OS/Arch:           windows/amd64
 Context:           desktop-linux

Server: Docker Desktop 4.63.0 (220185)
 Engine:
  Version:          29.2.1
  API version:      1.53 (minimum version 1.44)
  Go version:       go1.25.6
  Git commit:       6bc6209
  Built:            Mon Feb  2 17:17:24 2026
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v2.2.1
  GitCommit:        dea7da592f5d1d2b7755e3a161be07f43fad8f75
 runc:
  Version:          1.3.4
  GitCommit:        v1.3.4-0-gd6d73eb8
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
(base) PS C:\Users\L15> cd my-multi-app
(base) PS C:\Users\L15\my-multi-app> docker compose up -d --build
time="2026-07-16T19:41:46+05:30" level=warning msg="C:\\Users\\L15\\my-multi-app\\docker-compose.yml:
the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
"
#1 [internal] load local bake definitions
#1 reading from stdin 517B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 186B done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/python:3.9-slim
#3 DONE 2.6s

#4 [internal] load .dockerignore
```

```
#4 [internal] load .dockerignore
#4 transferring context: 2B done
#4 DONE 0.0s

#5 [internal] load build context
#5 transferring context: 582B done
#5 DONE 0.0s

#6 [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1
acda1e13cf1731b1b
#6 resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd
a1e13cf1731b1b 0.0s done
#6 sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 0B / 251B 0.2s
#6 sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 251B / 251B 0.3s done
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 0B / 13.88MB 0.2s
#6 sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 0B / 1.29MB 0.2s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 0B / 29.78MB 0.2s
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 1.05MB / 13.88MB 0.6s
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 3.15MB / 13.88MB 0.8s
#6 sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 1.29MB / 1.29MB 0.9s done
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 8.39MB / 13.88MB 1.1s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 2.10MB / 29.78MB 0.9s
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 10.49MB / 13.88MB 1.2s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 4.19MB / 29.78MB 1.1s
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 11.53MB / 13.88MB 1.4s
```

```
#6 extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 0.1s done
#6 extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08
#6 extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 0.3s done
#6 DONE 3.8s

#6 [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1
acda1e13cf1731b1b
#6 extracting sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 0.0s done
#6 DONE 3.8s

#7 [2/5] WORKDIR /app
#7 DONE 0.1s

#8 [3/5] COPY requirements.txt .
#8 DONE 0.0s

#9 [4/5] RUN pip install -r requirements.txt
#9 1.146 Collecting flask
#9 1.206   Downloading flask-3.1.3-py3-none-any.whl (103 kB)
#9 1.235      ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 103.4/103.4 kB 3.5 MB/s eta 0:00:00
#9 1.257 Collecting pymysql
#9 1.261   Downloading pymysql-1.2.0-py3-none-any.whl (45 kB)
#9 1.267      ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 45.7/45.7 kB 11.0 MB/s eta 0:00:00
#9 1.298 Collecting click>=8.1.3
#9 1.303   Downloading click-8.1.8-py3-none-any.whl (98 kB)
#9 1.311      ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 98.2/98.2 kB 14.7 MB/s eta 0:00:00
#9 1.400 Collecting markupsafe>=2.1.1
#9 1.405   Downloading markupsafe-3.0.3-cp39-cp39-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux
_2_28_x86_64.whl (20 kB)
#9 1.451 Collecting werkzeug>=3.1.0
#9 1.455   Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
#9 1.474      ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 226.5/226.5 kB 13.1 MB/s eta 0:00:00
#9 1.495 Collecting jinja2>=3.1.2
#9 1.499   Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
#9 1.512      ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 134.9/134.9 kB 11.8 MB/s eta 0:00:00
#9 1.528 Collecting itsdangerous>=2.2.0
#9 1.532   Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
#9 1.550 Collecting blinker>=1.9.0
#9 1.554   Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
#9 1.607 Collecting importlib-metadata>=3.6.0
#9 1.612   Downloading importlib_metadata-8.7.1-py3-none-any.whl (27 kB)
#9 1.659 Collecting zipp>=3.20
#9 1.665   Downloading zipp-3.23.1-py3-none-any.whl (10 kB)
#9 1.748 Installing collected packages: zipp, pymysql, markupsafe, itsdangerous, click, blinker, werkz
eug, jinja2, importlib-metadata, flask
#9 2.037 Successfully installed blinker-1.9.0 click-8.1.8 flask-3.1.3 importlib-metadata-8.7.1 itsdang
```

```
#9 2.037 Successfully installed blinker-1.9.0 click-8.1.8 flask-3.1.3 importlib-metadata-8.7.1 itsdang
erous-2.2.0 jinja2-3.1.6 markupsafe-3.0.3 pymysql-1.2.0 werkzeug-3.1.8 zipp-3.23.1
#9 2.037 WARNING: Running pip as the 'root' user can result in broken permissions and conflicting beha
viour with the system package manager. It is recommended to use a virtual environment instead: https:/
/pip.pypa.io/warnings/venv
#9 2.103
#9 2.103 [notice] A new release of pip is available: 23.0.1 -> 26.0.1
#9 2.103 [notice] To update, run: pip install --upgrade pip
#9 DONE 2.2s

#10 [5/5] COPY . .
#10 DONE 0.0s

#11 exporting to image
#11 exporting layers
#11 exporting layers 0.4s done
#11 exporting manifest sha256:c93c08f52d8f8849a6269f3a53ac922d052c8794be6358f7d8455f00013db295 done
#11 exporting config sha256:0d45ad071a03ee74f0e34cceef604bae8d05d8adc82bea9c1eff4ed64ae6655d done
#11 exporting attestation manifest sha256:c32edaa3e7aa84f40a4fb9f9faae90330bb12b3d0258ab9ac008e365b441
5f80 0.0s done
#11 exporting manifest list sha256:5a7d0d22545022b20578ae00521ae87eb859bcf6dd072cc2b97503a6f89a667d do
ne
#11 naming to docker.io/library/my-multi-app-web:latest done
#11 unpacking to docker.io/library/my-multi-app-web:latest
#11 unpacking to docker.io/library/my-multi-app-web:latest 0.2s done
#11 DONE 0.7s

#12 resolving provenance for metadata file
#12 DONE 0.0s
[+] up 5/5
 ✔Image my-multi-app-web       Built                                                    10.0s
 ✔Network my-multi-app_app-net Created                                                  0.0s
 ✔Volume my-multi-app_db_data  Created                                                  0.0s
 ✔Container my-multi-app-db-1  Created                                                  0.1s
 ✔Container my-multi-app-web-1 Created                                                  0.0s
```

```
(base) PS C:\Users\L15\my-multi-app> docker compose logs eb
time="2026-07-16T19:44:01+05:30" level=warning msg="C:\\Users\\L15\\my-multi-app\\docker-compose.yml:
the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
"
no such service: eb
(base) PS C:\Users\L15\my-multi-app> docker compose logs web
time="2026-07-16T19:44:06+05:30" level=warning msg="C:\\Users\\L15\\my-multi-app\\docker-compose.yml:
the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
"
web-1  |  * Serving Flask app 'app'
web-1  |  * Debug mode: off
web-1  | WARNING: This is a development server. Do not use it in a production deployment. Use a produc
tion WSGI server instead.
web-1  |  * Running on all addresses (0.0.0.0)
web-1  |  * Running on http://127.0.0.1:5000
web-1  |  * Running on http://172.18.0.3:5000
web-1  | Press CTRL+C to quit
web-1  | [2026-07-28 05:07:46,619] ERROR in app: Exception on / [GET]
web-1  | Traceback (most recent call last):
web-1  |   File "/usr/local/lib/python3.9/site-packages/flask/app.py", line 1511, in wsgi_app
web-1  |     response = self.full_dispatch_request()
web-1  |   File "/usr/local/lib/python3.9/site-packages/flask/app.py", line 919, in full_dispatch_requ
est
web-1  |     rv = self.handle_user_exception(e)
web-1  |   File "/usr/local/lib/python3.9/site-packages/flask/app.py", line 917, in full_dispatch_requ
est
web-1  |     rv = self.dispatch_request()
web-1  |   File "/usr/local/lib/python3.9/site-packages/flask/app.py", line 902, in dispatch_request
web-1  |     return self.ensure_sync(self.view_functions[rule.endpoint])(**view_args)  # type: ignore[
no-any-return]
web-1  |   File "/app/app.py", line 8, in hello
web-1  |     db_host = os.envirom.get('DB_HOST', 'localhost')
web-1  | AttributeError: module 'os' has no attribute 'envirom'
web-1  | 172.18.0.1 - - [28/Jul/2026 05:07:46] "GET / HTTP/1.1" 500 -
web-1  | 172.18.0.1 - - [28/Jul/2026 05:07:46] "GET /favicon.ico HTTP/1.1" 404 -
(base) PS C:\Users\L15\my-multi-app> docker compose restart web
time="2026-07-16T19:44:29+05:30" level=warning msg="C:\\Users\\L15\\my-multi-app\\docker-compose.yml:
the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
"
[+] restart 0/1
 - Container my-multi-app-web-1 Restarting                                                    1.6s
(base) PS C:\Users\L15\my-multi-app> docker compose up -d --build
time="2026-07-16T19:46:03+05:30" level=warning msg="C:\\Users\\L15\\my-multi-app\\docker-compose.yml:
the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
"
#1 [internal] load local bake definitions
#1 reading from stdin 517B done
#1 DONE 0.0s
```

```
#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 186B done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/python:3.9-slim
#3 DONE 0.7s

#4 [internal] load .dockerignore
#4 transferring context: 2B done
#4 DONE 0.0s

#5 [internal] load build context
#5 transferring context: 402B done
#5 DONE 0.0s

#6 [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1
acda1e13cf1731b1b
#6 resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd
a1e13cf1731b1b 0.0s done
#6 DONE 0.0s

#7 [2/5] WORKDIR /app
#7 CACHED

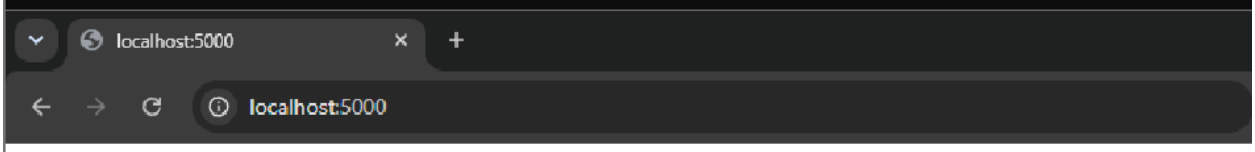
#8 [3/5] COPY requirements.txt .
#8 CACHED

#9 [4/5] RUN pip install -r requirements.txt
#9 CACHED

#10 [5/5] COPY . .
#10 DONE 0.0s

#11 exporting to image
#11 exporting layers 0.0s done
#11 exporting manifest sha256:697190bad60844df5696f2453dd35257527f63a96ee18264c38a4ff1c5ab46aa done
#11 exporting config sha256:1663bce7e96603fbf5b87844b122c62b17b40635bd7c3985f3fa57bfd21d4b0a
#11 exporting config sha256:1663bce7e96603fbf5b87844b122c62b17b40635bd7c3985f3fa57bfd21d4b0a done
#11 exporting attestation manifest sha256:5f590dcf015946408ae5b97292bd6e1fbf36d06b0f9321cf381f065743c8
8dd8 0.0s done
#11 exporting manifest list sha256:5e99d2763b172a89e35087349109c798f4b72414badb388c214bf3a6787fda27 do
ne
#11 naming to docker.io/library/my-multi-app-web:latest done
#11 unpacking to docker.io/library/my-multi-app-web:latest 0.0s done
#11 DONE 0.1s

#12 resolving provenance for metadata file
#12 DONE 0.0s
[+] up 3/3
 ✔Image my-multi-app-web       Built                                    1.3s
 ✔Container my-multi-app-db-1  Running                                  0.0s
 ✔Container my-multi-app-web-1 Recreated                                1.4s
(base) PS C:\Users\L15\my-multi-app>
```



Hello from Docker Compose! Connected to database host: db